HW2 Assessment Criteria

	Excellent	Good	Satisfactory	Unsatisfactory
Functionality (20 pt.)	The design performs its specified function perfectly, passing more than 90% of the test cases. (**18-20 pt.**)	The design mostly performs its specified function correctly, passing more than 75% of the test cases. **(15-17 pt.)**	The design partially performs its specified function correctly, passing more than 50% of the test cases. **(10-14 pt.)**	The design only partially fulfills its specified function, passing less than 50% of the test cases. **(0-9 pt.)**
Verilog Syntax (5 pt.)	Correctly implement your algorithm and gate-level circuit structure. And the syntax is hardware-oriented encoding. **(5 pt.)**	Correctly implement the algorithm and gate-level circuit structure. But only part of the syntax is hardware-oriented encoding. **(3-4 pt.)**	Correctly implement your algorithm and gate-level circuit structure. But the syntax is software-oriented encoding. **(1-2 pt.)**	Only partially implement your algorithm and gate-level circuit structure, the correctness is unsatisfied. **(0 pt.)**
Algorithm (15 pt.)	Use multiple algorithms for comprehensive comparison and evaluation. Clearly explain the characteristics, advantages, disadvantage, operation methods, etc. of these algorithms. **(12-15 pt.)**	Use one kind of algorithm. Clearly explain its characteristics, advantages, disadvantages, operation methods, etc. **(8-11 pt.)**	Use one kind of algorithm. Only explain part of its characteristics, advantages, disadvantages, operation methods, etc, not complete enough. **(4-7 pt.)**	Use one kind algorithm. Briefly describe its characteristics, advantages, disadvantages, operation methods, etc, not complete enough and poor structure. **(0-3 pt.)**
Architecture (10 pt)	Use multiple architecture for comprehensive comparison and evaluation. Clearly explain the characteristics,	Use one kind of architecture. Clearly explain its characteristics, advantages, disadvantages, operation methods,	Use one kind of architecture. Only explain part of its characteristics, advantages, disadvantages, operation methods,	Use one kind algorithm. Briefly describe its characteristics, advantages, disadvantages, operation methods,

	advantages, disadvantage, operation methods, etc. of these algorithms. **(9-10 pt.)**	etc. **(6-8 pt.)**	etc, not complete enough. **(3-5 pt.)**	etc, not complete enough and poor structure. **(0-2 pt.)**
Circuit Analysis (15 pt.)	Use multiple architectures for comprehensive comparisons and evaluations. Clearly analyze the number of logic gates, the number of logic gate layers, the critical path indication, etc. of these algorithms. **(12-15 pt.)**	Use one kind of architectures. Clearly explain the logic elements (adder, multiplexer, etc.), the critical path indication, etc. of this algorithms. **(8-11 pt.)**	Use one kind of architectures. Only explain part of the logic elements (adder, multiplexer, etc.), the critical path indication, etc. of this algorithms. Description is not complete enough. **(4-7 pt.)**	Use one kind of architectures. Briefly explain part of the logic elements (adder, multiplexer, etc.), the critical path indication, etc. of this algorithms. Not complete enough and poor structure. **(0-3 pt.)**
Testbench and completeness of verification (15 pt.)	The verification and waveforms are clear and complete, considering various corner cases. The verification signal generation and comparison in testbench are performed automatically. **(14-15 pt.)**	The verification and waveforms are clear and complete, considering various corner cases. However, the verification signal generation and comparison in testbench are performed manually. **(11-13 pt.)**	Complete the basic verification, perform the waveform. However only consider part of corner cases. **(7-10 pt.)**	Only partial verification and wave pattern display are performed. The completeness is not good. **(0-6 pt.)**
Discussion (15 pt.)	Clearly explain the problems and solutions encountered during the design process, And reflect on the impact on the design in the future. **(14-15 pt.)**	Clearly explain the problems encountered during the design process and their solutions, but there is no reflection. **(11-13 pt.)**	It only describes some of the problems encountered during the design process and solutions. Not complete enough. **(7-10 pt.)**	Only briefly state the problems encountered during the design process and solutions. Poor structure. **(0-6 pt.)**

Report Overall (10 pt.)	Reports are clear and easy to read. Fonts, font styles, line spacing, indentations, pictures, etc. are neat and unified, and the paragraph hierarchy is clear and distinct. **(9-10 pt.)**	Reports are clear and easy to read. Fonts, font styles, line spacing, indentations, pictures, etc. are neat and unified, However, the paragraph hierarchy is no clear enough. **(6-8 pt.)**	Only some fonts, font styles, line spacing, indentations, pictures, etc. of the report are neat. The paragraph hierarchy is not complete enough. **(3-5 pt.)**	The fonts, font styles, line spacing, indentations, pictures and other formats in the report are disordered, and the paragraph hierarchical structure is poor. **(0-2 pt.)**